

Title

Engagement Patterns and Therapeutic Mediators in Digital Anxiety Intervention

Abstract

We conducted post-hoc mediation and subgroup analyses on data from our 96-participant trial comparing a digital anxiety intervention plus standard care to standard care alone. Participants with higher baseline anxiety severity (top quartile) showed greater absolute symptom reductions (mean -3.4 vs -1.8 in lower-severity group, $p = 0.018$). Mediation modeling suggested that approximately 40% of the intervention's effect on anxiety was mediated through increases in reported behavioral engagement and social re-engagement. Participants achieving at least weekly engagement with core application features showed 2.3-fold greater likelihood of meeting the study's clinical response criterion. Subgroup analysis by age revealed larger effect sizes in participants under age 50 compared to older adults ($d = 0.68$ vs 0.42). Acceptability ratings were similar across demographic subgroups. These findings highlight that intervention efficacy depends partly on engagement frequency and suggest that tailored engagement strategies may optimize outcomes in older or initially lower-engagement populations.